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IMAGE FORMING APPARATUS

Abstract

An image forming apparatus comprising an apparatus body including a first developing unit including a first developing container and a first developing roller, and a second developing unit including a second developing container and a second developing roller, a first toner container configured to attachably and detachably supported on the apparatus body, the first toner container being configured to supply the first toner together with air toward the first developing container, a second toner container configured to attachably and detachably supported on the apparatus body, the second toner container being configured to supply the second toner together with air toward the second developing container, a pump unit including a first pump and a second pump, and aligned with the first pump in the first direction, and one driving source configured to drive the first pump and second pump.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an image forming apparatus including a pump unit.

Description of the Related Art

[0002] Hitherto, an image forming apparatus in which is installed a cartridge equipped with a developing container that accommodates toner and an attachment portion capable of having a supply pack accommodating toner attached thereto is proposed (refer to Japanese Patent Application Laid-Open Publication No. 2023-134401). Toner discharged from the supply pack attached to the attachment portion is accommodated temporarily in a reserve tank disposed on a cartridge, and conveyed from the reserve tank to the developing container. Toner is conveyed from the reserve tank to the developing container by a screw or air flowing from a pump.

SUMMARY OF THE INVENTION

[0003] According to a first aspect of the present invention, an image forming apparatus configured to form an image on a sheet, comprising an apparatus body including a first developing unit including a first developing container configured to accommodate a first toner, and a first developing roller configured to bear the first toner, and a second developing unit including a second developing container configured to accommodate a second toner, and a second developing roller configured to bear the second toner, a first toner container configured to accommodate the first toner and attachably and detachably supported on the apparatus body, the first toner container being configured to supply the first toner together with air toward the first developing container, a second toner container configured to accommodate the second toner and attachably and detachably supported on the apparatus body, the second toner container being configured to supply the second toner together with air toward the second developing container, the second toner container being aligned in a first direction with the first toner container, a pump unit including a first pump configured to convey air toward the first toner container, and a second pump configured to convey air toward the second toner container, and aligned with the first pump in the first direction; and one driving source configured to drive the first pump and second pump.

[0004] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is an entire schematic view of an image forming apparatus according to a present embodiment.

[0006] FIG. 2A is a perspective view of the image forming apparatus.

[0007] FIG. 2B is a perspective view of the image forming apparatus with a front door opened.

[0008] FIG. 2C is a perspective view of the image forming apparatus with a toner cartridge removed.

[0009] FIG. 3A is a perspective view of an image forming unit.

[0010] FIG. 3B is a perspective view of the image forming unit with the toner cartridge removed.

[0011] FIG. 4A is a plan view of a laser scanner and the image forming unit.

[0012] FIG. 4B is a plan view of the image forming unit.
[0013] FIG. 5A is a cross-sectional view taken at a 5A-5A cross-section of FIG. 4A.
[0014] FIG. 5B is a cross-sectional view taken at a 5B-5B cross-section of FIG. 4A.
[0015] FIG. 6 is a perspective view of a pump unit.
[0016] FIG. 7 is an exploded perspective view of the pump unit.
[0017] FIG. 8A is a front view of the toner cartridge.
[0018] FIG. 8B is a plan view of the toner cartridge.
[0019] FIG. 8C is a bottom view of the toner cartridge.
[0020] FIG. 8D is a side view of the toner cartridge.
[0021] FIG. 8E is a rear view of the toner cartridge.
[0022] FIG. 9A is a cross-sectional view taken at a 9A-9A cross-section of FIG. 8E.
[0023] FIG. 9B is a perspective view of the toner cartridge cut at an XZ plane.
[0024] FIG. 9C is an exploded view illustrating the toner cartridge cut at the XZ plane.
[0025] FIG. 10A is a cross-sectional view taken at a 10A-10A cross-section of FIG. 8B.
[0026] FIG. 10B is a perspective view of the toner cartridge cut at a YZ plane.

DESCRIPTION OF THE EMBODIMENTS

Entire Configuration

[0027] An image forming apparatus **100** according to the present embodiment is a laser beam printer that adopts an electrophotographic system. The image forming apparatus refers to an apparatus for forming an image on a sheet used as a recording medium based on an image information entered from an external PC or an image information read from a document, and which includes a printer, a copying machine, a facsimile, and a multifunction machine. Further, the image forming apparatus may have an auxiliary device such as an optional feeder, an image reading apparatus, or a sheet processing apparatus additionally connected to a main body having an image forming function, and the entire system having such an auxiliary device connected thereto is also one kind of an image forming apparatus.

[0028] The image forming apparatus **100** includes, as illustrated in FIG. 1, an image forming portion **40** for forming an image on a sheet S, a sheet feed unit **18**, a fixing unit **21**, and a sheet discharge roller pair **22**. The image forming portion **40** includes four process cartridges PY, PM, PC, and PK for forming toner images of four colors, which are yellow (Y), magenta (M), cyan (C), and black (K), and a laser scanner LB. The laser scanner LB serving as an exposure unit is arranged above the process cartridges PY, PM, PC, and PK. Alternatively, an LED exposure unit may be adopted instead of the laser scanner LB.

[0029] The portion excluding the process cartridges PY, PM, PC, and PK and toner cartridges **430Y**, **430M**, **430C**, and **430K** described later from the image forming apparatus **100** may be referred to as a main body or an apparatus body **72** of the image forming apparatus **100**. In the following description, the toner cartridges **430Y**, **430M**, **430C**, and **430K** may simply be referred to as cartridges **430Y**, **430M**, **430C**, and **430K**. Further, the toner cartridge **430K** is an example of a first toner container capable of supplying black toner together with air to a developing container **43K**, and the toner cartridge **430C** is an example of a second toner container capable of supplying cyan toner together with air to a developing container **43C**. The process cartridges PY, PM, PC, and PK may be attachably and detachably supported on the apparatus body **72** or may be fixed in a non-detachable manner to the apparatus body **72**.

[0030] In the present embodiment, suffixes P, Y, M, C, and K are attached to reference numbers of components corresponding to the toner colors of yellow, magenta, cyan, and black, respectively. The configurations and operations of components to which the suffixes are attached are approximately the same, except for the different toner colors. Therefore, in the following description, unless distinction is specifically necessary, descriptions may be provided in a general manner with the suffixes Y to K omitted.

[0031] Further, in the following description, unless denoted otherwise, directions X, Y, and Z are

defined as follows, assuming that the process cartridges PY, PM, PC, and PK and the toner cartridges **430Y**, **430M**, **430C**, and **430K** take their normal positions, that is, the positions that they take when being attached to the apparatus body **72**.

[0032] As illustrated in FIG. **2A**, a front-rear direction is denoted by an X axis, and a direction from a rear side toward a front side of the image forming apparatus **100** is referred to as an X direction. The X direction may also be referred to as a second direction, a forward direction, a front direction, or an attachment/detachment direction of the toner cartridge. Further, a downstream side in the X direction of the image forming apparatus **100** may be referred to as a front side, and an upstream side thereof may be referred to as a rear side.

[0033] A right-left direction is denoted by a Y axis, and a direction from left to right of the image forming apparatus **100** is referred to as a Y direction. The Y direction may also be referred to as a direction of rotational axis **AL1**, a first direction, and a right direction. Further, a downstream side in the Y direction of the image forming apparatus **100** may be referred to as a right side, and the upstream side thereof may be referred to as a left side.

[0034] An up-down direction is denoted by a Z axis, and a direction from a lower side toward an upper side of the image forming apparatus **100** is referred to as a Z direction. The Z direction may also be referred to as a third direction, an upper direction, a height direction, and a vertical direction. Further, a downstream side in the Z direction of the image forming apparatus **100** may be referred to as an upper side, an upper surface side, and a top surface side, whereas an upstream side thereof may be referred to as a lower side, a lower surface side, and a bottom surface side.

[0035] The X axis, the Y axis, and the Z axis are mutually perpendicular to each other. For example, the X axis is perpendicular to the Y axis, and also perpendicular to the Z axis. A plane perpendicular to the X axis may be referred to as a YZ plane, a plane perpendicular to the Y axis may be referred to as a ZX plane, and a plane perpendicular to the Z axis may be referred to as a XY plane. For example, the XY plane is a horizontal plane. The X direction and the Y direction are directions along the horizontal XY plane, that is, the horizontal direction.

[0036] The process cartridge PY includes a drum unit **8Y** and a developing unit **9Y**. The drum unit **8Y** includes a photosensitive drum **44Y** and a charging roller **45Y**. The photosensitive drum **44Y** is formed by applying an organic photoconductive layer on an outer circumference of an aluminum cylinder, and is rotated by a drive motor not shown. A photosensitive belt may be used instead of the photosensitive drum **44Y**. The developing unit **9Y** includes a developing container **43Y** that accommodates yellow toner, and a developing roller **46Y** that develops an electrostatic latent image on the photosensitive drum **44Y**.

[0037] Similarly, the process cartridge PM includes a drum unit **8M** and a developing unit **9M**. The drum unit **8M** includes a photosensitive drum **44M** and a charging roller **45M**. The photosensitive drum **44M** is formed by applying an organic photoconductive layer on an outer circumference of an aluminum cylinder, and is rotated by a drive motor not shown. A photosensitive belt may be used instead of the photosensitive drum **44M**. The developing unit **9M** includes a developing container **43M** that accommodates magenta toner, and a developing roller **46M** that develops an electrostatic latent image on the photosensitive drum **44M**.

[0038] Similarly, the process cartridge PC includes a drum unit **8C** and a developing unit **9C**. The drum unit **8C** includes a photosensitive drum **44C** serving as a second photosensitive drum and a charging roller **45C**. The photosensitive drum **44C** is formed by applying an organic photoconductive layer on an outer circumference of an aluminum cylinder, and is rotated by a drive motor not shown. A photosensitive belt may be used instead of the photosensitive drum **44C**. The developing unit **9C** serving as a second developing unit includes a developing container **43C** that accommodates cyan toner, and a developing roller **46C** that develops an electrostatic latent image on the photosensitive drum **44C**. That is, the developing container **43C** serving as a second developing container is configured to accommodate cyan toner serving as a second toner, and the developing roller **46C** serving as a second developing roller is configured to bear cyan toner. The

developing units **9Y** and **9M** may each be assumed as the second developing unit.

[0039] Similarly, the process cartridge PK includes a drum unit **8K** and a developing unit **9K**. The drum unit **8K** includes a photosensitive drum **44K** serving as a first photosensitive drum, and a charging roller **45K**. The photosensitive drum **44K** is formed by applying an organic photoconductive layer on an outer circumference of an aluminum cylinder, and is rotated by a drive motor not shown. A photosensitive belt may be used instead of the photosensitive drum **44K**. The developing unit **9K** serving as a first developing unit includes a developing container **43K** that accommodates black toner, and a developing roller **46K** that develops an electrostatic latent image on the photosensitive drum **44K**. That is, the developing container **43K** serving as a first developing container is configured to accommodate black toner serving as a first toner, and the developing roller **46K** serving as a first developing roller is configured to bear black toner.

[0040] These four process cartridges PY, PM, PC, and PK are arranged in an aligned manner in the X direction. Further, an intermediate transfer belt unit **11** is disposed below the process cartridges PY, PM, PC, and PK in the image forming portion **40**. The intermediate transfer belt unit **11** includes an intermediate transfer belt **12** wound around a drive roller **14**, a tension roller **13**, and an assist roller **15**, and a secondary transfer roller **17**, and the intermediate transfer belt **12** is rotated by the drive roller **14** in a clockwise direction in FIG. **1**. Primary transfer rollers **16Y**, **16M**, **16C**, and **16K** are disposed on an inner side of the intermediate transfer belt **12**.

[0041] The primary transfer rollers **16Y**, **16M**, **16C**, and **16K** each come into contact with the photosensitive drums **44Y**, **44M**, **44C**, and **44K**, thereby forming primary transfer portions **30Y**, **30M**, **30C**, and **30K**. The secondary transfer roller **17** is arranged to face the drive roller **14** with the intermediate transfer belt **12** interposed therebetween. The secondary transfer roller **17** and the intermediate transfer belt **12** come into contact with each other and form a secondary transfer portion **31**.

[0042] The fixing unit **21** includes a fixing film **21a** heated by a heater, and a pressure roller **21b** that comes into pressure contact with the fixing film **21a**. The sheet feed unit **18** is disposed on a lower portion of the image forming apparatus **100**. The sheet feed unit **18** includes a feeding cassette **19** that may be drawn out of and attached to the apparatus body **72**, and a feed roller **20** that feeds sheets S accommodated in the feeding cassette **19**. Further, the sheet feed unit **18** includes a manual sheet feed port **330** through which sheets are manually inserted. Sheets may include paper such as normal paper and envelopes, plastic films as overhead projector (OHP) sheets, and cloths.

Image Forming Operation

[0043] Next, an image forming operation of the image forming apparatus **100** configured in this manner will be described. When an image signal is entered to the laser scanner LB from a personal computer not shown, laser lights LY, LM, LC, and LK corresponding to the image signal are irradiated from the laser scanner LB. The laser lights LY, LM, LC, and LK for exposing the respective photosensitive drums are irradiated on the photosensitive drums **44Y**, **44M**, **44C**, and **44K** through exposure windows **10Y**, **10M**, **10C**, and **10K** of the laser scanner LB.

[0044] In this state, a surface of the photosensitive drum **44Y** is charged uniformly in advance to a predetermined polarity and potential by the charging roller **45Y**, and an electrostatic latent image is formed on the surface by being irradiated with a laser light LY from the laser scanner LB. The electrostatic latent image formed on the photosensitive drum **44Y** is developed by the developing roller **46Y**, and a yellow (Y) toner image is formed on the photosensitive drum **44Y**.

[0045] Similarly, magenta (M), cyan (C), and black (K) toner images are formed respectively on the photosensitive drums **44M**, **44C**, and **44K**. The toner images of respective colors formed on the respective photosensitive drums are transferred to the intermediate transfer belt **12** by the primary transfer rollers **16Y**, **16M**, **16C**, and **16K**, and conveyed on the intermediate transfer belt **12** being rotated by the drive roller **14** to the secondary transfer roller **17**.

[0046] The image forming processes of the respective colors are performed at a timing such that

each toner image is superposed on a toner image formed upstream therefrom and primarily transferred to the intermediate transfer belt **12**. Further, after the toner images of the respective colors have been transferred to the intermediate transfer belt **12**, toner remaining on the surface of the respective photosensitive drums are removed by a cleaning device not shown.

[0047] In parallel with the above-mentioned image forming process, the sheet **S** accommodated in the feeding cassette **19** of the sheet feed unit **18** or the sheet supplied from the manual sheet feed port **330** are conveyed toward the secondary transfer portion **31**. In the secondary transfer portion **31**, a full-color toner image on the intermediate transfer belt **12** is transferred to the sheet **S** by a secondary transfer bias applied to the secondary transfer roller **17**. Predetermined heat and pressure is applied from the fixing film **21a** and the pressure roller **21b** of the fixing unit **21** to the sheet **S** to which the toner image has been transferred, by which the toner is melted and fixed. The sheet **S** having passed through the fixing unit **21** is discharged in a direction along the **X** direction by the sheet discharge roller pair **22**, and supported on a sheet discharge tray **23** disposed on an upper portion of the apparatus body **72**.

General Configuration of Toner Cartridge

[0048] Next, a general configuration of the toner cartridges **430Y**, **430M**, **430C**, and **430K** will be described with reference to FIGS. **2A** to **2C**. FIG. **2A** is a perspective view of the image forming apparatus **100**, FIG. **2B** is a perspective view of the image forming apparatus **100** in a state where a front door **72b** is opened, and FIG. **2C** is a perspective view of the image forming apparatus **100** in a state where the toner cartridge **430Y** is removed.

[0049] As illustrated in FIG. **2A**, the image forming apparatus **100** includes the front door **72b** that is supported in an openable/closable manner on the casing of the apparatus body **72**. The front door **72b** covers an opening **72a** provided on a front portion, that is, a downstream end portion in the **X** direction, of the apparatus body **72** by being positioned in a closed position, as illustrated in FIGS. **2A** and **2B**. Further, the front door **72b** opens the opening **72a** of the apparatus body **72** by being positioned at an open position, as illustrated in FIG. **2B**. The front door **72b** may maintain its position at the open position.

[0050] As illustrated in FIG. **2B**, when the front door **72b** is opened from the closed position to the open position, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are exposed to the exterior of the image forming apparatus **100** through the opening **72a**. Therefore, the user may access the toner cartridges **430Y**, **430M**, **430C**, and **430K**. The toner cartridges **430Y**, **430M**, **430C**, and **430K** may supply toner to the developing containers **43Y**, **43M**, **43C**, and **43K** of the process cartridges **PY**, **PM**, **PC**, and **PK**.

[0051] The toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged downstream in the **X** direction and downstream in the **Z** direction of the apparatus body **72**, that is, on the front side and upper side of the apparatus body **72**. In other words, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged downstream of the apparatus body **72** in a sheet discharge direction of the sheet discharge roller pair **22**. The toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged in an aligned manner in the **Y** direction as an alignment direction. That is, the **X** direction, which is the direction of alignment of the process cartridges **PY**, **PM**, **PC**, and **PK**, and the **Y** direction, which is the direction of alignment of the toner cartridges **430Y**, **430M**, **430C**, and **430K**, are mutually orthogonal.

[0052] As illustrated in FIG. **2C**, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are attachably and detachably supported in the **X** direction with respect to a cartridge holder **51** disposed on the apparatus body **72**. That is, an **X** direction, and similarly the -**X** direction, serving as a second direction intersecting the **Y** direction is the attachment/detachment direction of the toner cartridges **430Y**, **430M**, **430C**, and **430K**. Therefore, the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be replaced without having to remove the process cartridges **PY**, **PM**, **PC**, and **PK** from the apparatus body **72**. The toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged on a front side of the apparatus body **72** and are exposed by opening the front door **72b**, such that the cartridges

may be replaced easily.

[0053] The cartridge holder **51** is arranged on the front side of the apparatus body **72** and downstream of the process cartridges PY, PM, PC, and PK in the X direction. Therefore, even in a case where the toner cartridges **430Y**, **430M**, **430C**, and **430K** are removed from the cartridge holder **51**, the process cartridges PY, PM, PC, and PK are covered by the cartridge holder **51** and are not exposed to the exterior of the image forming apparatus **100** through the opening **72a**. The toner cartridges **430Y**, **430M**, **430C**, and **430K** are stored in the apparatus body **72** in a state where the front door **72b** is positioned at the closed position.

[0054] Further, as illustrated in FIG. 2A, indicators **208Y**, **208M**, **208C**, and **208K** are disposed on a front side **72c** of the apparatus body **72**. The indicators **208Y**, **208M**, **208C**, and **208K** are for indicating the colors of toners accommodated in each of the toner cartridges **430Y**, **430M**, **430C**, and **430K** to the user in an easily recognizable manner. The indicators **208Y**, **208M**, **208C**, and **208K** may be composed of LED or stickers, and for example, the indicator **208Y** may be composed of an LED that is lit or flashed in yellow, or a sticker colored in yellow. Further, the indicators **208Y**, **208M**, **208C**, and **208K** may be provided with a function to display a remaining amount of toner accommodated in each of the corresponding process cartridges PY, PM, PC, and PK.

[0055] As illustrated in FIG. 2B, an inclined portion **51a** is formed on a lower portion of the cartridge holder **51**. The inclined portion **51a** is an inclined surface facing an oblique upper direction. In other words, the inclined portion **51a** is an inclined surface that extends downstream in the Z direction toward an upstream side in the X direction. Labels **210Y**, **210M**, **210C**, and **210K** corresponding to toner colors accommodated in the toner cartridges **430Y**, **430M**, **430C**, and **430K** are attached to the inclined portion **51a**. For example, the labels **210Y**, **210M**, **210C**, and **210K** may be colored in yellow, magenta, cyan, and black, or have the letters yellow, magenta, cyan, and black printed thereon. The user may visually confirm the labels **210Y**, **210M**, **210C**, and **210K** attached to the inclined portion **51a** when replacing the toner cartridges **430Y**, **430M**, **430C**, and **430K**. Therefore, the user may attach the respective toner cartridges to the appropriate positions.

Image Forming Unit

[0056] Next, an image forming unit **500** including a toner conveyance mechanism for conveying toner to the process cartridges PY, PM, PC, and PK from the toner cartridges **430Y**, **430M**, **430C**, and **430K** will be described with reference to FIGS. 3A to 5B. FIG. 3A is a perspective view of the image forming unit **500**, and FIG. 3B is a perspective view of the image forming unit **500** in a state where the toner cartridges **430Y**, **430M**, **430C**, and **430K** are removed.

[0057] FIG. 4A is a plan view of the laser scanner LB and the image forming unit **500**, and FIG. 4B is a plan view of the image forming unit **500**. FIG. 5A is a cross-sectional view showing a 5A-5A cross-section of FIG. 4A, and FIG. 5B is a cross-sectional view showing a 5B-5B cross-section of FIG. 4A.

[0058] The image forming unit **500** includes, as illustrated in FIGS. 3A and 3B, the process cartridges PY, PM, PC, and PK, the toner cartridges **430Y**, **430M**, **430C**, and **430K**, the cartridge holder **51**, and a pump unit **1**. The pump unit **1** is arranged on a rear direction of the cartridge holder **51** and an upper direction of the process cartridge PK, as illustrated in FIGS. 1 and 3A.

[0059] As illustrated in FIG. 3B, air conveyed from the pump unit **1** is discharged through discharge holes **514a**, **514b**, **514c**, and **514d** disposed on the cartridge holder **51**. The discharge holes **514a**, **514b**, **514c**, and **514d** are opened upward, that is, in the Z direction.

[0060] Further, inlet ports **51aY**, **51aM**, **51aC**, and **51aK** are formed on the cartridge holder **51**. The inlet ports **51aY**, **51aM**, **51aC**, and **51aK** are through holes formed on a rear surface of the cartridge holder **51**. The opening direction of the inlet ports **51aY**, **51aM**, **51aC**, and **51aK** are the X direction, such that they intersect with the Z direction, which is the opening direction of the discharge holes **514a**, **514b**, **514c**, and **514d**.

[0061] In a state where the toner cartridges **430Y**, **430M**, **430C**, and **430K** are attached to the cartridge holder **51**, air discharged through the discharge holes **514a**, **514b**, **514c**, and **514d** are

supplied to the interior of the toner cartridges **430Y**, **430M**, **430C**, and **430K**. Then, toner discharged together with air from the toner cartridges **430Y**, **430M**, **430C**, and **430K** are conveyed through the inlet ports **51aY**, **51aM**, **51aC**, and **51aK** of the cartridge holder **51** to supply pipes **62Y**, **62M**, **62C**, and **62K**.

[0062] As illustrated in FIG. **4B**, the supply pipes **62Y**, **62M**, **62C**, and **62K** include upstream end portions **62Yu**, **62Mu**, **62Cu**, and **62Ku** respectively connected to the inlet ports **51aY**, **51aM**, **51aC**, and **51aK**, and downstream end portions **62Yd**, **62Md**, **62Cd**, and **62Kd** respectively connected to the developing containers **43Y**, **43M**, **43C**, and **43K**.

[0063] Then, toner discharged together with air from the toner cartridges **430Y**, **430M**, **430C**, and **430K** are passed through the supply pipes **62Y**, **62M**, **62C**, and **62K** and supplied to the developing containers **43Y**, **43M**, **43C**, and **43K**.

[0064] The supply pipes **62Y** and **62M** are respectively routed on an upstream end side in the Y direction of the developing containers **43Y** and **3M**. The supply pipes **62C** and **62K** are respectively routed on a downstream end side in the Y direction of developing containers **43C** and **3K**. As described, by separating the supply pipes **62Y**, **62M**, **62C**, and **62K** to be respectively routed on either the left side or the right side of the developing containers **43Y**, **43M**, **43C**, and **43K**, the lengths of the supply pipes **62Y**, **62M**, **62C**, and **62K** may be shortened. Therefore, pressure loss in the supply pipes **62Y**, **62M**, **62C**, and **62K** may be reduced, and the image forming unit **500** may be downsized.

[0065] Next, an arrangement of the toner cartridge **430Y** will be described. As illustrated in FIG. **5B**, the toner cartridge **430Y** and the process cartridges PY, PM, PC, and PK are arranged to overlap with a virtual line VL1 that extends in the X direction. That is, as illustrated in FIGS. **1** and **5B**, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged in an aligned manner with the process cartridges PY, PM, PC, and PK in the X direction. In other words, when viewed in the X direction, at least a portion of each of the toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged to be overlapped with the process cartridges PY, PM, PC, and PK in the Z direction. Therefore, the image forming unit **500** may be downsized in the Z direction, that is, in the up-down direction, and the image forming apparatus **100** may be downsized.

[0066] In the present embodiment, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged as described above, but the present technique is not limited thereto. That is, the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be arranged above the process cartridges PY, PM, PC, and PK in the Z direction.

[0067] Further, the image forming unit **500** should preferably adopt a configuration that allows easy access to the process cartridges PY, PM, PC, and PK and the supply pipes **62Y**, **62M**, **62C**, and **62K** so as to realize maintenance and replacement of the process cartridges PY, PM, PC, and PK and the supply pipes **62Y**, **62M**, **62C**, and **62K**. Therefore, according to the present embodiment, the image forming unit **500** includes a tray **55** that supports the process cartridges PY, PM, PC, and PK and the cartridge holder **51**. The tray **55** is configured to be drawn out in the X direction from the apparatus body **72**. The user may easily access the process cartridges PY, PM, PC, and PK, the toner cartridges **430Y**, **430M**, **430C**, and **430K** supported by the cartridge holder **51**, and the pump unit **1** by drawing out the tray **55** in the X direction in a state where the front door **72b** is opened.

Pump Unit

[0068] Next, the pump unit **1** will be described with reference to FIGS. **6** and **7**. FIG. **6** is a perspective view illustrating the pump unit **1** according to the present embodiment. FIG. **7** is a exploded perspective view of the pump unit **1**. The pump unit **1** illustrated in FIGS. **6** and **7** is positioned in a state where the pump unit **1** illustrated in FIGS. **1** to **5B** is rotated by **180** degrees about the X axis. In other words, a short direction of the pump unit **1** is denoted by the X axis, and a direction from the rear side toward the front side of the pump unit **1** is referred to as the X direction. A rotation shaft direction of a first dual shaft motor **4L** and a second dual shaft motor **4R** of the pump unit **1** is denoted by the Y axis, and a direction from the second dual shaft motor **4R**

toward the first dual shaft motor **4L** is referred to as the Y direction. The Y direction is sometimes referred to as an axial direction or a longitudinal direction. An up-down direction of the pump unit **1** is denoted by the Z axis, and the Z direction may be referred to as a height direction or a vertical direction.

[0069] The pump unit **1** includes, as illustrated in FIGS. **6** and **7**, the first dual shaft motor **4L**, the second dual shaft motor **4R**, a first drive transmission portion **5L1**, a second drive transmission portion **5L2**, a third drive transmission portion **5R1**, and a fourth drive transmission portion **5R2**. The pump unit **1** includes a first pump **3L1**, a second pump **3L2**, a third pump **3R1**, a fourth pump **3R2**, a first holding member **7L**, a second holding member **7R**, a first pipe **6L1**, a second pipe **6L2**, a third pipe **6R1**, and a fourth pipe **6R2**.

[0070] The first pump **3L1** conveys, or sends out, air to the first pipe **6L1** by having the drive of the first dual shaft motor **4L** transmitted via the first drive transmission portion **5L1**. The first pipe **6L1** is connected to a flow path not shown disposed within the cartridge holder **51**. The air sent to the first pipe **6L1** is conveyed through the flow path to the discharge hole **514d**. The discharge hole **514d** is connected to an intake port of the toner cartridge **430K** described below, and air is supplied through the discharge hole **514d** to the toner cartridge **430K**.

[0071] The second pump **3L2** conveys air to the second pipe **6L2** by having the drive of the first dual shaft motor **4L** transmitted via the second drive transmission portion **5L2**. The second pipe **6L2** is connected to a flow path not shown disposed within the cartridge holder **51**. The air sent to the second pipe **6L2** is conveyed through the flow path to the discharge hole **514c**. The discharge hole **514c** is connected to an intake port of the toner cartridge **430C** described below, and air is supplied through the discharge hole **514d** to the toner cartridge **430C**.

[0072] The third pump **3R1** conveys air to the third pipe **6R1** by having the drive of the second dual shaft motor **4R** transmitted via the third drive transmission portion **5R1**. The third pipe **6R1** is connected to a flow path not shown disposed within the cartridge holder **51**. The air sent to the third pipe **6R1** is conveyed through the flow path to the discharge hole **514b**. The discharge hole **514b** is connected to an intake port of the toner cartridge **430M** described below, and air is supplied through the discharge hole **514b** to the toner cartridge **430M**.

[0073] The fourth pump **3R2** conveys air to the fourth pipe **6R2** by having the drive of the second dual shaft motor **4R** transmitted via the fourth drive transmission portion **5R2**. The fourth pipe **6R2** is connected to a flow path not shown disposed within the cartridge holder **51**. The air sent to the fourth pipe **6R2** is conveyed through the flow path to the discharge hole **514a**. The discharge hole **514a** is connected to an intake port **430Yb1** of the toner cartridge **430Y** described below, and air is supplied through the discharge hole **514b** to the toner cartridge **430Y**.

[0074] The respective flow paths disposed in the cartridge holder **51** are formed independently within the cartridge holder **51**, and are not mutually communicated with each other.

[0075] The pump unit **1** includes an attachment member **37**, and the first dual shaft motor **4L** is held by the first holding member **7L** attached to the attachment member **37**. The second dual shaft motor **4R** is held by the second holding member **7R** attached to the attachment member **37**.

[0076] The first dual shaft motor **4L** serving as a motor includes a first rotation shaft **4Ls** serving as a rotation shaft that extends in the Y direction, and a first casing **4Lc** that rotatably supports the first rotation shaft **4Ls**. The first rotation shaft **4Ls** passes through the first casing **4Lc**, and is configured rotatably about a rotational axis **AL1** in a first rotation direction **R1** and a second rotation direction **R2** opposite to the first rotation direction **R1**. Further, the first rotation shaft **4Ls** includes a first projection portion **4La** that projects from one side of the first casing **4Lc** in the Y direction, and a second projection portion **4Lb** that projects from the other side of the first casing **4Lc**. The first projection portion **4La** serving as a first end portion is disposed on one side of the first rotation shaft **4Ls** in the Y direction, and a second projection portion **4Lb** serving as a second end portion is disposed on the other side of the first rotation shaft **4Ls** in the Y direction.

[0077] The second dual shaft motor **4R** includes a second rotation shaft **4Rs** that extends in the Y

direction, and a second casing 4Rc that rotatably supports the second rotation shaft 4Rs. The second rotation shaft 4Rs passes through the second casing 4Rc, and is configured rotatably about the rotational axis AL1 in a third rotation direction R3 and a fourth rotation direction R4 opposite to the third rotation direction R3. Further, the second rotation shaft 4Rs includes a third projection portion 4Ra that protrudes from one side of the second casing 4Rc in the Y direction, and a fourth projection portion 4Rb that protrudes from the other side the second casing 4Rc. The third projection portion 4Ra serving as a third end portion is disposed on one side of the second rotation shaft 4Rs in the Y direction, and the fourth projection portion 4Rb serving as a fourth end portion is disposed on the other side of the second rotation shaft 4Rs in the Y direction.

[0078] The first drive transmission portion 5L1 and the second drive transmission portion 5L2 are respectively attached to the first projection portion 4La and the second projection portion 4Lb of the first dual shaft motor 4L. The first drive transmission portion 5L1 includes, as illustrated in FIG. 7, a first clutch portion 5dL1, and a first eccentric shaft 5eL1 disposed on an output member 5c of the first clutch portion 5dL1. The first clutch portion 5dL1 includes an input member 5a to which drive from the first projection portion 4La is entered, the output member 5c, and a clutch member 5b disposed between the input member 5a and the output member 5c. The clutch member 5b is driven to rotate following the rotation of the input member 5a that rotates together with the first projection portion 4La. The clutch member 5b transmits drive to or disconnects the same from the output member 5c.

[0079] The clutch member 5b has a saw-toothed surface, and transmits drive to the output member 5c only when the first rotation shaft 4Ls of the first dual shaft motor 4L rotates in the first rotation direction R1. That is, the first clutch portion 5dL1 is set to be in a transmission state when the first rotation shaft 4Ls rotates in the first rotation direction R1, and set to be in a non-transmission state where drive is not transmitted when the first rotation shaft 4Ls rotates in the second rotation direction R2.

[0080] The first eccentric shaft 5eL1 disposed on the output member 5c is provided eccentrically with respect to the first rotation shaft 4Ls, and rotates integrally with the output member 5c that rotates about the first rotation shaft 4Ls. The first eccentric shaft 5eL1 is connected to the first pump 3L1, and by the first eccentric shaft 5eL1 rotating in the first rotation direction R1 about the first rotation shaft 4Ls, the first pump 3L1 operates. By the operation of the first pump 3L1, air is discharged through a first discharge portion 35L1 of the first pump 3L1. The first pipe 6L1 is connected to the first discharge portion 35L1, as illustrated in FIG. 6, and air discharged through the first discharge portion 35L1 is passed through the first pipe 6L1 and sent toward the toner cartridge 430K.

[0081] Similarly, the second drive transmission portion 5L2 includes, as illustrated in FIGS. 6 and 7, a second clutch portion, and a second eccentric shaft 5eL2 disposed on an output member of the second clutch portion. The second clutch portion has a similar configuration as the first clutch portion 5dL1, and it is arranged symmetrically in the Y direction with respect to the first casing 4Lc from the first clutch portion 5dL1.

[0082] The second eccentric shaft 5eL2 disposed on the output member of the second clutch portion is provided eccentrically with respect to the first rotation shaft 4Ls, and rotates integrally with the output member that rotates about the first rotation shaft 4Ls. The second eccentric shaft 5eL2 is connected to the second pump 3L2, and by the second eccentric shaft 5eL2 rotating in the second rotation direction R2 about the first rotation shaft 4Ls, the second pump 3L2 operates. By the operation of the second pump 3L2, air is discharged through a second discharge portion 35L2 of the second pump 3L2. The second pipe 6L2 is connected to the second discharge portion 35L2, as illustrated in FIG. 6, and air discharged through the second discharge portion 35L2 is passed through the second pipe 6L2 and sent toward the toner cartridge 430C.

[0083] The third drive transmission portion 5R1 and the fourth drive transmission portion 5R2 are respectively attached to the third projection portion 4Ra and the fourth projection portion 4Rb of

the second dual shaft motor **4R**. The third drive transmission portion **5R1** includes, as illustrated in FIGS. **6** and **7**, a third clutch portion, and a third eccentric shaft **5eR1** disposed on an output member of the third clutch portion. The third clutch portion adopts a similar configuration as the first clutch portion **5dL1**.

[0084] The third eccentric shaft **5eR1** disposed on the output member of the third clutch portion is provided eccentrically with respect to the second rotation shaft **4Rs**, and rotates integrally with the output member that rotates about the second rotation shaft **4Rs**. The third eccentric shaft **5eR1** is connected to the third pump **3R1**, and by the third eccentric shaft **5eR1** rotating in the third rotation direction **R3** about the second rotation shaft **4Rs**, the third pump **3R1** operates. By the operation of the third pump **3R1**, air is discharged through a third discharge portion **35R1** of the third pump **3R1**. The third pipe **6R1** is connected to the third discharge portion **35R1**, as illustrated in FIG. **6**, and air discharged through the third discharge portion **35R1** is passed through the third pipe **6R1** and sent toward the toner cartridge **430M**.

[0085] Similarly, the fourth drive transmission portion **5R2** includes, as illustrated in FIGS. **6** and **7**, a fourth clutch portion, and a fourth eccentric shaft **5eR2** disposed on an output member of the fourth clutch portion. The fourth clutch portion has a similar configuration as the first clutch portion **5dL1**, and it is arranged symmetrically in the Y direction with respect to the second casing **4Rc** from the third clutch portion.

[0086] The fourth eccentric shaft **5eR2** disposed on the output member of the fourth clutch portion is provided eccentrically with respect to the second rotation shaft **4Rs**, and rotates integrally with the output member that rotates about the second rotation shaft **4Rs**. The fourth eccentric shaft **5eR2** is connected to the fourth pump **3R2**, and by the fourth eccentric shaft **5eR2** rotating in the fourth rotation direction **R4** about the second rotation shaft **4Rs**, the fourth pump **3R2** operates. By the operation of the fourth pump **3R2**, air is discharged through a fourth discharge portion **35R2** of the fourth pump **3R2**. The fourth pipe **6R2** is connected to the fourth discharge portion **35R2**, as illustrated in FIG. **6**, and air discharged through the fourth discharge portion **35R2** is passed through the fourth pipe **6R2** and sent toward the toner cartridge **430Y**.

[0087] As described above, the respective clutch portions adopt similar configurations, and the clutch portions may be transitioned between a transmission state and a non-transmission state according to the rotation direction of the first rotation shaft **4Ls** and the rotation direction of the second rotation shaft **4Rs**.

[0088] That is, when the first rotation shaft **4Ls** of the first dual shaft motor **4L** rotates in the first rotation direction **R1**, the first clutch portion **5dL1** will be in a transmission state and the second clutch portion will be in a non-transmission state. Therefore, the first pump **3L1** will be driven by being transmitted a first driving force from a first projection portion **4La** of the first rotation shaft **4Ls** and the second pump **3L2** will not be driven. Further, when the first rotation shaft **4Ls** of the first dual shaft motor **4L** rotates in the second rotation direction **R2**, the first clutch portion **5dL1** will be in a non-transmission state and the second clutch portion will be in a transmission state. Therefore, the first pump **3L1** will not be driven and the second pump **3L2** will be driven by being transmitted a second driving force from the second projection portion **4Ra** of the first rotation shaft **4Ls**.

[0089] Further, when a second rotation shaft **4Rs** of a second dual shaft motor **4R** rotates in the third rotation direction **R3**, the third clutch portion will be in a transmission state and the fourth clutch portion will be in a non-transmission state. Therefore, the third pump **3R1** will be driven and the fourth pump **3R2** will not be driven. Further, as described above, when the second rotation shaft **4Rs** of the second dual shaft motor **4R** rotates in the fourth rotation direction **R4**, the third clutch portion will be in a non-transmission state and the fourth clutch portion will be in a transmission state. Therefore, the third pump **3R1** will not be driven and the fourth pump **3R2** will be driven.

[0090] Transmission states of the first clutch portion **5dL1**, the second clutch portion, the third clutch portion, and the fourth clutch portion may also be referred to respectively as a first

transmission state, a second transmission state, a third transmission state, and a fourth transmission state. The states of the first drive transmission portion **5L1**, the second drive transmission portion **5L2**, the third drive transmission portion **5R1**, and the fourth drive transmission portion **5R2** at this time may also be referred to respectively as the first transmission state, the second transmission state, the third transmission state, and the fourth transmission state.

[0091] The non-transmission states of the first clutch portion **5dL1**, the second clutch portion, the third clutch portion, and the fourth clutch portion may also be referred to respectively as a first non-transmission state, a second non-transmission state, a third non-transmission state, and a fourth non-transmission state. Further, the states of the first drive transmission portion **5L1**, the second drive transmission portion **5L2**, the third drive transmission portion **5R1**, and the fourth drive transmission portion **5R2** at this time may also be referred to respectively as the first non-transmission state, the second non-transmission state, the third non-transmission state, and the fourth non-transmission state.

[0092] The configurations of the respective clutch portions described above are mere examples, and the present technique is not limited thereto. That is, the clutch portion may adopt other configurations as long as the cutoff or transmission of drive to the output shaft is varied according to the rotation direction of the rotation shaft.

Detailed Configuration of Pump

[0093] Next, detailed configurations of the respective pumps will be described with reference to FIG. 7. The respective pumps adopt similar configurations as described above, such that in the following description, the first pump **3L1** is taken as an example and illustrated. As illustrated in FIG. 7, the first pump **3L1** includes a base member **36L1** fixed to the attachment member **37**, and a diaphragm member **31L1** nipped by the attachment member **37** and the base member **36L1**. The base member **36L1** includes an intake portion **34** through which air is taken in, a discharge portion **35L1** through which air is discharged, and a hole portion not shown.

[0094] The diaphragm member **31L1** includes a flat surface portion **31b**, and a pillar portion **31bL1** that extends in the Z direction from the flat surface portion **31b**. When the base member **36L1** and the diaphragm member **31L1** are attached to the attachment member **37**, the pillar portion **31bL1** of the diaphragm member **31L1** passes through the hole portion of the base member **36L1**. A suction valve **31c** and an exhaust valve **31d** are disposed on the flat surface portion **31b**. A shaft hole portion **31aL1** to which the first eccentric shaft **5eL1** is inserted via a bearing **32** is disposed on the pillar portion **31bL1**. In other words, the first eccentric shaft **5eL1** is supported rotatably in the shaft hole portion **31aL1** via the bearing **32**.

[0095] A recess portion **37a** is disposed on the attachment member **37** at a position corresponding to the flat surface portion **31b** of the diaphragm member **31L1**. In a state where the diaphragm member **31L1** and the base member **36L1** are attached to the attachment member **37**, a volume variation portion **33L1** is formed by the recess portion **37a** of the attachment member **37** and the flat surface portion **31b** of the diaphragm member **31L1**. The volume variation portion **33L1** is a space surrounded by the recess portion **37a** and the flat surface portion **31b**, and the first pump **3L1** is a diaphragm-type pump capable of varying the capacity of the volume variation portion **33L1**, which is the internal space of the first pump **3L1**. The capacity of the volume variation portion **33L1** repeatedly increases and decreases by the diaphragm member **31L1** being moved in response to the rotation of the first eccentric shaft **5eL1**. Thereby, the first pump **3L1** functions as a diaphragm-type pump that takes in air through the intake portion **34** and discharges air through the discharge portion **35L1**.

[0096] The second pump **3L2**, the third pump **3R1**, and the fourth pump **3R2** adopt a similar configuration as the first pump **3L1**, and may send out air by the rotation of the respective eccentric shafts. As described, the pump unit **1** may operate arbitrary pumps (**3L1**, **3L2**, **3R1**, and **3R2**) by switching the rotation directions of the rotation shafts (**4Ls** and **4Rs**) of the two motors, which are the first dual shaft motor **4L** and the second dual shaft motor **4R**.

Configuration of Toner Cartridge

[0097] Next, the general configuration of the toner cartridge **430Y** will be described with reference to FIGS. **8A** to **10B**. FIG. **8A** is a front view of the toner cartridge **430Y**, and FIG. **8B** is a plan view of the toner cartridge **430Y**. FIG. **8C** is a bottom view of the toner cartridge **430Y**, FIG. **8D** is a side view of the toner cartridge **430Y**, and FIG. **8E** is a rear view of the toner cartridge **430Y**. FIG. **9A** is a cross-sectional view of a **9A-9A** cross-section of FIG. **8E**, and FIG. **9B** is a perspective view of the toner cartridge **430Y** cut at an XZ plane. FIG. **9C** is an exploded view of the toner cartridge **430Y** cut at an XZ plane. FIG. **10A** is a cross-sectional view of a **10A-10A** cross-section of FIG. **8B**, and FIG. **10B** is a perspective view of the toner cartridge **430Y** cut at a YZ plane.

[0098] The toner cartridge **430K** has a width L_a in the Y direction, as illustrated in FIG. **3A**, and the width L_a is wider than each width L_{b1} of the toner cartridges **430Y**, **430M**, and **430C**.

Therefore, a capacity of the toner cartridge **430K** for accommodating toner is greater than the capacities of the respective toner cartridges **430Y**, **430M**, and **430C**. Generally, a consumption amount of black toner is greater than that of other color toners, and as described above, the toner cartridge **430K** may accommodate more toner than the other toner cartridges **430Y**, **430M**, and **430C**. Therefore, the number of times of replacement of the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be equalized, and the usability may be improved.

[0099] The toner cartridges **430Y**, **430M**, **430C**, and **430K** have similar configurations except for the widths in the Y direction, such that only the toner cartridge **430Y** will be described below, and the descriptions of the toner cartridges **430M**, **430C**, and **430K** will be omitted. The following descriptions related to the toner cartridge **430Y** are based on a position in which the toner cartridge **430Y** is attached to the cartridge holder **51**.

[0100] The toner cartridge **430Y** includes, as illustrated in FIGS. **8A** to **10B**, a first frame body **430Ya**, a second frame body **430Yb**, a filter **83Y**, and an air discharge duct **85Y**. According to the present embodiment, the first frame body **430Ya** and the second frame body **430Yb** are composed of a resin material, but they may also be formed of paper or rubber. The filter **83Y** is formed in a sheet shape, and fixed in a manner nipped between the first frame body **430Ya** and the second frame body **430Yb**.

[0101] An inner space SPY of the toner cartridge **430Y** is partitioned by the filter **83Y** into a toner accommodation chamber **430Yc** and an air chamber **430Yd**. That is, the toner accommodation chamber **430Yc** is composed of the first frame body **430Ya** and the filter **83Y**, and the air chamber **430Yd** is composed of the second frame body **430Yb** and the filter **83Y**. The air chamber **430Yd** is arranged below the filter **83Y**, and the toner accommodation chamber **430Yc** is arranged above the filter **83Y**.

[0102] The toner accommodation chamber **430Yc** is configured to accommodate toner T (refer to FIG. **9A**), and toner T is supported in a gravity direction, i.e., $-Z$ direction, by the filter **83Y** within the toner accommodation chamber **430Yc**. Toner T is not accommodated in the air chamber **430Yd**. The filter **83Y** is composed, for example, of a porous member formed of resin fiber. The pores of the filter **83Y** are designed to have a size and density that allows air to pass through but regulates toner T from passing through. That is, the filter **83Y** is configured to allow passing of air and prevent passing of toner T.

[0103] A flange portion **430Ya2** that extends along an XY plane is formed at a lower end portion of the first frame body **430Ya**, and a flange portion **430Yb3** that extends along the XY plane is formed on an upper end portion of the second frame body **430Yb**. The first frame body **430Ya** and the second frame body **430Yb** are joined by ultrasonic welding of the flange portions **430Ya2** and **430Yb3**, and an outer edge portion **83Ya** of the filter **83Y** is nipped between the first frame body **430Ya** and the second frame body **430Yb**. The flange portions **430Ya2** and **430Yb3** may be mutually fixed to each other, not only through ultrasonic welding, but through heat sealing, adhesive agents such as double-sided tape or hot melt, or by screws.

[0104] The filter **83Y** is inclined from the outer edge portion **83Ya** toward a lowermost portion

83Yb positioned below the outer edge portion **83Ya**. That is, the filter **83Y** includes an inclined portion that is disposed downward toward, that is, as it approximates, the lowermost portion **83Yb** in the X direction or the Y direction or the horizontal direction. The lowermost portion **83Yb** is a portion that is protruded with respect to the outer edge portion **83Ya** in a direction from the toner accommodation chamber **430Yc** toward the air chamber **430Yd**. The lowermost portion **83Yb** is disposed at a center portion of the filter **83Y** in the X direction and the Y direction, as illustrated in FIGS. **9A** and **10B**.

[0105] As illustrated in FIG. **8E**, a discharge port **430Ya1** through which toner in the toner accommodation chamber **430Yc** is discharged to the exterior of the toner cartridge **430Y** is disposed on a rear surface **4300Ya** of the first frame body **430Ya**. The discharge port **430Ya1** is composed of a through hole that passes through the first frame body **430Ya** in the X direction.

[0106] As illustrated in FIGS. **8A**, **8C**, and **8E**, a projected portion **4301** is disposed to protrude downward on a bottom surface **4300Yb** of the second frame body **430Yb**. The intake port **430Yb1** composed of a through port that passes through the second frame body **430Yb** in the Z direction is disposed on the projected portion **4301**. The discharge port **430Ya1** and the intake port **430Yb1** are communicated with the exterior of the toner cartridge **430Y**.

[0107] The discharge port **430Ya1** and the intake port **430Yb1** are preferably disposed on sides constituting the toner cartridge **430Y** excluding the right and left sides thereof. In other words, the discharge port **430Ya1** and the intake port **430Yb1** are preferably disposed on the side surfaces of the toner cartridge **430Y** excluding the side facing the Y direction, which is the direction along which the toner cartridges **430Y**, **430M**, **430C**, and **430K** are aligned. Thereby, as illustrated in FIG. **3A**, gaps **Gym**, **Gmc**, and **Gck** formed between the respective toner cartridges **430Y**, **430M**, **430C**, and **430K** in the Y direction may be made narrow. As a result, the widths of the toner cartridges **430Y**, **430M**, **430C**, and **430K** in the Y direction may be widened, and the capacity of toner capable of being accommodated in the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be increased.

[0108] The gap **Gym** is the gap formed between the toner cartridge **430Y** and the toner cartridge **430M** in the Y direction. The gap **Gmc** is the gap formed between the toner cartridge **430M** and the toner cartridge **430C** in the Y direction. The gap **Gck** is the gap formed between the toner cartridge **430C** and the toner cartridge **430K** in the Y direction.

[0109] As illustrated in FIG. **3A**, a positioning portion **51b** that fits to the projected portion **4301** is disposed on the cartridge holder **51**. In the present embodiment, the positioning portion **51b** is a recess portion formed on the cartridge holder **51**, and on the bottom surface of the positioning portion **51b** is formed the discharge hole **514a** through which air sent from the pump unit **1** is discharged. The toner cartridge **430Y** is positioned with respect to the cartridge holder **51** by having the projected portion **4301** fit to the positioning portion **51b**. In this state, the intake port **430Yb1** disposed on the projected portion **4301** of the toner cartridge **430Y** is communicated with the fourth pipe **6R2** of a pump unit **80Y**.

[0110] The discharge port **430Ya1** formed on the first frame body **430Ya** of the toner cartridge **430Y** is opened toward an upstream direction in the X direction. That is, the discharge port **430Ya1** is opened toward a same direction as an attachment direction of the toner cartridge **430Y** to the cartridge holder **51**. Therefore, when attaching the toner cartridge **430Y** to the cartridge holder **51** facing the upstream side in the X direction, the discharge port **430Ya1** may be easily fit to communicate with an inlet port **51aY** of the cartridge holder **51**.

[0111] Regardless of the arrangement described above, the discharge port **430Ya1** may be disposed on the bottom surface **4300Yb** or the upper surface of the toner cartridge **430Y**, and the intake port **430Yb1** may be disposed on the rear surface **4300Ya** or the upper surface of the toner cartridge **430Y**. Further, if there is enough space within the apparatus body **72**, the discharge port **430Ya1** and the intake port **430Yb1** may be disposed on both right and left sides among the respective sides of the toner cartridge **430Y**.

[0112] Further, the toner cartridge **430Y** may be provided with a sealing member not shown for

sealing the discharge port **430Ya1**. The sealing member is composed of a seal or a shutter. In a state where the toner cartridge **430Y** is not attached to the cartridge holder **51**, the sealing member may seal the discharge port **430Ya1** to prevent toner within the toner accommodation chamber **430Yc** from leaking to the exterior of the toner cartridge **430Y**. In a state where sealing member is composed of a seal, the sealing member will be removed from the rear surface **4300Ya** of the toner cartridge **430Y** when attaching the toner cartridge **430Y** to the cartridge holder **51**. In a state where the sealing member is composed of a shutter, the sealing member will be moved with respect to the rear surface **4300Ya** of the toner cartridge **430Y** to open the discharge port **430Ya1** when attaching the toner cartridge **430Y** to the cartridge holder **51**.

[0113] Further, as illustrated in FIG. **8A**, a label **430Ys** is provided on a front side **4304** of the toner cartridge **430Y**. The label **430Ys** is provided to indicate the color of toner accommodated in the toner cartridge **430Y**. The label **430Ys** is not shown in drawings other than FIG. **8A**. The label **430Ys** may also indicate the method for attaching the toner cartridge **430Y** to the cartridge holder **51** or information related to the toner cartridge **430Y**. For example, the label **430Ys** may indicate the capacity of toner that the toner cartridge **430Y** may accommodate, the expiration date for use of the toner, the method for storing the toner cartridge **430Y**, and the method for removing the sealing member described above.

[0114] As illustrated in FIGS. **9A** to **10B**, the air discharge duct **85Y** is arranged on the inner side of the toner accommodation chamber **430Yc**. The air discharge duct **85Y** includes an inlet **85Ya** that opens downward in the gravity direction, and an outlet **85Yb** that opens rearward, that is, upstream in the X direction, and communicates with the discharge port **430Ya1** of the toner cartridge **430Y** to be opened to the exterior of the toner cartridge **430Y**. The outlet **85Yb** is positioned upward than the inlet **85Ya**. The air discharge duct **85Y** includes a first pipe portion **85Y1** to which the inlet **85Ya** is provided and that extends in the Z direction, and a second pipe portion **85Y2** to which the outlet **85Yb** connected to the discharge port **430Ya1** is provided and that extends in the X direction, wherein the air discharge duct **85Y** is bent in midway. As described, the air discharge duct **85Y** connects the inlet **85Ya** and the outlet **85Yb**. The inlet **85Ya** is arranged to face the lowermost portion **83Yb** of the filter **83Y** with a gap formed therebetween in the gravity direction. The inlet **85Ya** is preferably positioned adjacent to the filter **83Y**. The air discharge duct **85Y** according to the present embodiment is composed of a resin material, but it may also be composed of paper or rubber.

Toner Conveyance Mechanism

[0115] Next, a mechanism by which toner accommodated in the toner accommodation chamber **430Yc** of the toner cartridge **430Y** is conveyed to the developing container **43Y** of the process cartridge PY will be described. As illustrated in FIGS. **3B** and **6**, by the operation of the fourth pump **3R2** of the pump unit **1**, air is discharged through the discharge hole **514a** via the flow path of the fourth pipe **6R2** and the cartridge holder **51**. Air discharged upward through the discharge hole **514a** is taken into the air chamber **430Yd** through the intake port **430Yb1** of the toner cartridge **430Y**, as illustrated in FIG. **9A**. The air is filled in the air chamber **430Yd**, and passes through the filter **83Y** into the toner accommodation chamber **430Yc**.

[0116] The air flowing into the toner accommodation chamber **430Yc** enters the space between particles of toner T and fluidifies the toner T. The interior of the toner cartridge **430Y** is set to positive pressure by air flowing in from the pump unit **80Y**, and the air attempts to exit via the air discharge duct **85Y** through the discharge port **430Ya1** to the exterior of the toner cartridge **430Y**. In this state, toner T within the toner accommodation chamber **430Yc** is caused to move within the air discharge duct **85Y** together with air, and is discharged to the exterior of the toner cartridge **430Y** through the discharge port **430Ya1**. That is, the air discharge duct **85Y** guides the toner within the toner accommodation chamber **430Yc** together with air through the inlet **85Ya** to the discharge port **430Ya1**.

[0117] Toner T discharged through the discharge port **430Ya1** is passed through the inlet port **51aY**

of the cartridge holder **51** and enters an upstream end portion **62Y_u** of the supply pipe **62Y**. Then, toner **T** is conveyed within the supply pipe **62Y** by the air that has flown into the supply pipe **62Y** together with toner **T**, and supplied through a downstream end portion **62Y_d** of the supply pipe **62Y** into the developing container **43Y** of the process cartridge **PY**.

[0118] Steering members **SY1** and **SY2** for steering toner within the developing container **43Y** is disposed rotatably in the interior of the developing container **43Y**, as illustrated in FIG. 5B.

Similarly, steering members **SM1** and **SM2**, steering members **SC1** and **SC2**, and steering members **SK1** and **SK2** are disposed respectively in the interior of the developing containers **43M**, **43C**, and **43K**.

[0119] Toner within the developing container **43Y** is leveled by the rotation of the steering members **SY1** and **SY2**, and conveyed in the **Y** direction. According to the present embodiment, the supply pipes **62Y**, **62M**, **62C**, and **62K** are respectively connected to the center portion in the **Y** direction of the developing units **9Y**, **9M**, **9C**, and **9K**, in other words, the developing containers **43Y**, **43M**, **43C**, and **43K**. Thereby, the time required for leveling the toner agent surface of respective colors by the rotation of the steering members **SY1** and **SY2** will not be dispersed in the respective developing units **9Y**, **9M**, **9C**, and **9K** corresponding to the four color toners. In the present embodiment, the steering members are composed of a rotation shaft and a sheet member fixed to the rotation shaft, but the present technique is not limited thereto. For example, the steering member may be composed of a screw that is configured to convey the toner within the developing container along the **Y** direction.

[0120] Not only toner **T** but also air flows into the developing container **43Y**, such that the internal pressure within the developing container **43Y** increases. Therefore, according to the present embodiment, as illustrated in FIGS. 3A and 4B, a through hole (not shown) is formed on the upper surface of the developing container **43Y**, and the through hole is covered with an air filter **PfY**. The air filter **PfY** is formed of nonwoven fabric, which allows air to pass through but regulates the passing of toner. Thereby, toner flowing into the developing container **43Y** is restricted by the air filter **PfY** from being discharged to the exterior, and remains inside the developing container **43Y**. Meanwhile, at least a portion of air flowing into the developing container **43Y** is passed through the air filter **PfY** and discharged to the exterior of the developing container **43Y**. Thereby, the increase of internal pressure of the developing container **43Y** is suppressed, and supply of toner to the developing container **43Y** may be realized smoothly.

[0121] Similarly, air filters **PfM**, **PfC**, and **PfK** and through holes not shown covered by the air filters **PfM**, **PfC**, and **PfK** are provided on the upper surfaces of the developing containers **43M**, **43C**, and **43K**.

[0122] The number of air filters and through holes are not limited to one, and multiple filters and through holes may be provided on the developing container **43Y**. By providing multiple air filters and through holes on one developing container, the increase of internal pressure within the developing container may be suppressed efficiently.

[0123] As described above, toner accommodated in the toner cartridge **430Y** is conveyed together with air to the developing container **43Y** and supplied to the developing container **43Y**. In a configuration where the direction of the conveyance path is changed in midway or if the conveyance paths differ according to toner color, as according to the supply pipes **62Y**, **62M**, **62C**, and **62K** of the present embodiment, it is preferable to convey toner using air. By conveying toner using air, the degree of freedom of design of the toner conveyance path is improved, and members such as screws for conveying toner become unnecessary, such that the number of necessary components may be reduced and costs may be cut down.

[0124] By providing the air chamber **430Y_d**, which is a space having high sealability, between the discharge hole **514a** and the filter **83Y** as according to the present embodiment, the air discharged through the discharge hole **514a** will head efficiently to the filter **83Y** without being dispersed to the exterior of the toner cartridge **430Y**. For example, in a situation where the toner **T** within the

toner accommodation chamber **430Yc** is aggregated, such as when vibration is applied to the toner cartridge **430Y** or when it has been left for a long period of time, the pressure necessary to allow air to flow into the toner accommodation chamber **430Yc** through the filter **83Y** becomes high. Even according to this case, by continuously sending air from the pump unit **1** to the air chamber **430Yd** and increasing the air, i.e., air pressure, within the air chamber **430Yd**, air may be sent into the toner accommodation chamber **430Yc** through the filter **83Y**.

[0125] Further, since the air chamber **430Yd** is provided, the pump unit **1** is merely required to exert a performance of creating a pressure to send air continuously into the air chamber **430Yd** until a necessary pressure is realized to allow air to pass through the filter **83Y** into the toner accommodation chamber **430Yc**. Since there are no special requirements regarding the discharge speed or discharge amount of air, it becomes necessary to adopt a small pump unit, such that the downsizing of the apparatus is realized.

Pump Unit Layout

[0126] Next, a layout of the pump unit **1** will be described in detail. As illustrated in FIGS. **6** and **7**, the first rotation shaft **4Ls** of the first dual shaft motor **4L** and the second rotation shaft **4Rs** of the second dual shaft motor **4R** are arranged coaxially. That is, the first rotation shaft **4Ls** and the second rotation shaft **4Rs** are arranged on the same rotational axis **AL1** as illustrated in FIG. **1**. Therefore, the size of the pump unit **1** in the X direction and the Z direction may be downsized.

[0127] The first dual shaft motor **4L**, the first pump **3L1**, the first drive transmission portion **5L1**, the second drive transmission portion **5L2**, and the second pump **3L2** are aligned in the Y direction. Similarly, the second dual shaft motor **4R**, the third pump **3R1**, the third drive transmission portion **5R1**, the fourth drive transmission portion **5R2**, and the fourth pump **3R2** are aligned in the Y direction. Further, the first pump **3L1**, the second pump **3L2**, the third pump **3R1**, and the fourth pump **3R2** are aligned in the Y direction serving as the first direction. Therefore, the size of the pump unit **1** in the X direction and the Z direction may be downsized.

[0128] As described above, by downsizing the pump unit **1** in the X direction and the Z direction, the pump unit **1** may be arranged between a laser light **LC** exposing the photosensitive drum **44C** and a laser light **LK** exposing the photosensitive drum **44K** in the X direction, as illustrated in FIG. **1**. The laser light **LC** is one example of a second light path for exposing the photosensitive drum **44C**, and the laser light **LK** is one example of a first light path for exposing the photosensitive drum **44K**. Further, the pump unit **1** may be arranged between the laser scanner **LB** and the developing unit **9K** in the Z direction, which serves as a third direction perpendicular to both the X direction and the Y direction. Especially, as illustrated in FIG. **5B**, in a case where the developing units **9Y**, **9M**, **9C**, and **9K** are arranged in a manner separating in the Z direction from the laser scanner **LB** along the X direction, a space is formed between the laser scanner **LB** and the developing unit **9K**. By arranging the pump unit **1** in this space, space efficiency of the image forming apparatus **100** may be improved.

[0129] The pump unit **1** is arranged at a position farther from the fixing unit **21** than the developing units **9Y**, **9M**, and **9C** are from the fixing unit **21** in the X direction. Further, the developing unit **9Y** is arranged closer to the fixing unit **21** than the developing units **9M**, **9C**, and **9K** are to the fixing unit **21** in the X direction, and the developing unit **9M** is arranged closer to the fixing unit **21** than the developing units **9C** and **9K** in the X direction. Further, the developing unit **9C** is arranged closer to the fixing unit **21** than the developing unit **9K** in the X direction. The fixing unit **21** generates heat in the process of performing a fixing operation of fixing the toner image on a sheet. Therefore, by arranging the pump unit **1** at a position distant from the fixing unit **21**, the rising of temperature of the first dual shaft motor **4L** and the second dual shaft motor **4R** of the pump unit **1** may be suppressed. Further, air flow may be created in the image forming unit **500** by the pump unit **1**, and temperature of air around the image forming unit **500** may be reduced.

[0130] The intake ports (**430Yb1**) of the respective cartridges through which air sent from the respective pumps are taken in are also arranged in an aligned manner in the Y direction. Therefore,

the path of air from the respective pumps, such as the first pump **3L1**, to the respective intake ports, such as **430Yb1**, may be shortened, such that the conveyance efficiency of air may be improved, and occupation area of the pump unit **1** and the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be reduced.

[0131] Further, as described above, the first pump **3L1** and the second pump **3L2** may be driven selectively by the first dual shaft motor **4L**, which is one of the driving sources. Similarly, the third pump **3R1** and the fourth pump **3R2** may be driven selectively by the second dual shaft motor **4R**, which is one of the driving sources. Therefore, the number of driving sources of the pump unit **1** may be reduced, by which space may be saved and costs may be cut down.

[0132] As illustrated in FIGS. **1** and **5B**, when viewed in the Y direction, the developing units **9Y**, **9M**, **9C**, and **9K** and the toner cartridges **430Y**, **430M**, **430C**, and **430K** are arranged to be overlapped with the virtual line **VL1** extending in the X direction. The pump unit **1** is arranged above the virtual line **VL1**. Further, the toner cartridges **430Y**, **430M**, **430C**, and **430K** are disposed closer to the developing unit **9K** than to the developing units **9Y**, **9M**, and **9C** in the X direction, and the pump unit **1** is disposed so as to be overlapped with the developing unit **9K** when viewed in the Z direction. Thereby, the pump unit **1** may be arranged close to the toner cartridges **430Y**, **430M**, **430C**, and **430K** and the developing unit **9K**. Thereby, pressure loss when sending air between the pump unit **1** and the toner cartridges **430Y**, **430M**, **430C**, and **430K** may be reduced, and the pump unit **1** and the image forming unit **500** may be downsized.

[0133] By adopting the arrangement of the pump unit **1** and the circumference configurations as described above, a new embodiment of the image forming apparatus including the pump unit **1** may be offered. Further, the pump unit **1** and the image forming apparatus **100** may be downsized.

Other Embodiments

[0134] According to the present embodiment, the first pump **3L1** is operated when the first rotation shaft **4Ls** is rotated in the first rotation direction **R1**, but the present technique is not limited thereto. Which pump is to be operated when the first rotation shaft **4Ls** or the second rotation shaft **4Rs** is rotated in the respective directions may be set arbitrarily.

[0135] According to the present embodiment, the first dual shaft motor **4L**, the first clutch portion **5dL1**, the first eccentric shaft **5eL1**, the second clutch portion, and the second eccentric shaft **5eL2** are arranged in an aligned manner in the Y direction, but the present technique is not limited thereto. For example, the first dual shaft motor **4L**, the first clutch portion **5dL1**, the first eccentric shaft **5eL1**, the second clutch portion, and the second eccentric shaft **5eL2** may be arranged in an aligned manner in either the X direction or the Z direction.

[0136] According to the present embodiment, the first rotation shaft **4Ls** of the first dual shaft motor **4L** and the second rotation shaft **4Rs** of the second dual shaft motor **4R** are arranged coaxially, but the present technique is not limited thereto. For example, the first rotation shaft **4Ls** and the second rotation shaft **4Rs** are not necessarily arranged coaxially, and may be somewhat displaced in the X direction or the Z direction.

[0137] The respective pumps described above are not limited to diaphragm-type pumps, and they may be composed of reciprocating pumps, such as piston pumps or plunger pumps. Further, the respective pumps may be composed of non-positive-displacement pumps such as centrifugal pumps, propeller pumps, or viscosity pumps. According to the pump unit **1**, the first pump **3L1** and the second pump **3L2** are operated by one first dual shaft motor **4L**, but the present technique is not limited thereto. For example, the first pump **3L1** and the second pump **3L2** may be operated not by one driving source but by two driving sources.

[0138] Further according to the present embodiment, the pump unit **1** is arranged above the developing unit **9K**, but the present technique is not limited thereto. For example, the pump unit **1** may be arranged below the developing unit **9K**. In this case, the pump unit **1** may be arranged so as not to overlap with the toner cartridges **430Y**, **430M**, **430C**, and **430K** when viewed in the Z direction.

[0139] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0140] This application claims the benefit of Japanese Patent Application No. 2024-027892, filed Feb. 27, 2024 which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus configured to form an image on a sheet, comprising: an apparatus body including: a first developing unit including a first developing container configured to accommodate a first toner and a first developing roller configured to bear the first toner; and a second developing unit including a second developing container configured to accommodate a second toner and a second developing roller configured to bear the second toner, a first toner container configured to accommodate the first toner and attachably and detachably supported on the apparatus body, the first toner container being configured to supply the first toner together with air toward the first developing container; a second toner container configured to accommodate the second toner and attachably and detachably supported on the apparatus body, the second toner container being configured to supply the second toner together with air toward the second developing container, the second toner container being aligned in a first direction with the first toner container; a pump unit including: a first pump configured to convey air toward the first toner container, and a second pump configured to convey air toward the second toner container, and aligned with the first pump in the first direction; and one driving source configured to drive the first pump and second pump.
2. The image forming apparatus according to claim 1, wherein the first developing unit and the second developing unit are aligned in a second direction orthogonal to the first direction.
3. The image forming apparatus according to claim 2, wherein the first toner container and the second toner container are disposed, in the second direction, closer to the first developing unit than to the second developing unit, and wherein the pump unit is disposed so as to be overlapped with the first developing unit when viewed in a third direction perpendicular to both the first direction and the second direction.
4. The image forming apparatus according to claim 3, wherein the pump unit is disposed above the first developing unit.
5. The image forming apparatus according to claim 1, wherein the pump unit includes: a motor configured to serve as the driving source, the motor including a rotation shaft configured to rotate in a first rotation direction and a second rotation direction opposite to the first rotation direction about a rotational axis, the rotation shaft including a first end portion on one side in a direction of the rotational axis and a second end portion on the other side thereof, a first drive transmission portion configured to be transitioned between a first transmission state in which the first drive transmission portion transmits a first driving force from the first end portion of the rotation shaft to the first pump, and a first non-transmission state in which the first drive transmission portion does not transmit the first driving force from the first end portion to the first pump; and a second drive transmission portion configured to be transitioned between a second transmission state in which the second drive transmission portion transmits a second driving force from the second end portion of the rotation shaft to the second pump, and a second non-transmission state in which the second drive transmission portion does not transmit the second driving force from the second end portion to the second pump, wherein the first drive transmission portion is configured to drive the first pump by being transitioned to the first transmission state when the rotation shaft is rotated in the first rotation direction, and configured not to drive the first pump by being transitioned to the first non-transmission state when the rotation shaft is rotated in the second rotation direction, and

- wherein the second drive transmission portion is configured not to drive the second pump by being transitioned to the second non-transmission state when the rotation shaft is rotated in the first rotation direction, and configured to drive the second pump by being transitioned to the second transmission state when the rotation shaft is rotated in the second rotation direction.
- 6.** The image forming apparatus according to claim 5, wherein the motor, the first pump, the second pump, the first drive transmission portion, and the second drive transmission portion are aligned in the first direction.
- 7.** The image forming apparatus according to claim 5, wherein each of the first pump and the second pump is a diaphragm-pump.
- 8.** The image forming apparatus according to claim 2, wherein the apparatus body includes a fixing unit configured to fix a toner image to the sheet by applying heat and pressure to the toner image, wherein the second developing unit is arranged closer to the fixing unit than the first developing unit is to the fixing unit in the second direction, and wherein the pump unit is arranged farther from the fixing unit than the second developing unit is from the fixing unit in the second direction.
- 9.** The image forming apparatus according to claim 1, wherein the first developing unit, the second developing unit, and the pump unit are aligned in a second direction orthogonal to the first direction.
- 10.** The image forming apparatus according to claim 2, wherein, when viewed in the first direction, the first developing unit, the second developing unit, the first toner container, and the second toner container are arranged so as to be overlapped with a virtual line that extends in the second direction, and wherein the pump unit is disposed above the virtual line.
- 11.** The image forming apparatus according to claim 2, wherein the apparatus body includes: a first photosensitive drum configured to bear an electrostatic latent image to be developed by the first developing roller, a second photosensitive drum configured to bear an electrostatic latent image to be developed by the second developing roller; and an exposure unit configured to expose the first photosensitive drum and the second photosensitive drum, and wherein the pump unit is arranged, in the second direction, between a first light path by which the exposure unit exposes the first photosensitive drum and a second light path by which the exposure unit exposes the second photosensitive drum.
- 12.** The image forming apparatus according to claim 2, wherein the apparatus body includes: a first photosensitive drum configured to bear an electrostatic latent image developed by the first developing roller, a second photosensitive drum configured to bear an electrostatic latent image developed by the second developing roller; and an exposure unit configured to expose the first photosensitive drum and the second photosensitive drum, and wherein the pump unit is disposed between the exposure unit and the first developing unit in a third direction perpendicular to both the first direction and the second direction.
- 13.** An image forming apparatus configured to form an image on a sheet, comprising: an apparatus body including: a first developing unit including a first developing container configured to accommodate a first toner, and a first developing roller configured to bear the first toner; and a second developing unit including a second developing container configured to accommodate a second toner, and a second developing roller configured to bear the second toner, a first toner container configured to accommodate the first toner and attachably and detachably supported on the apparatus body, the first toner container being configured to supply the first toner together with air toward the first developing container; a second toner container configured to accommodate the second toner and supported attachably and detachably on the apparatus body, the second toner container being configured to supply the second toner together with air toward the second developing container, the second toner container being aligned in a first direction with the first toner container; and a pump unit including: a first pump configured to convey air toward the first toner container, and a second pump configured to convey air toward the second toner container, wherein the first developing unit and the second developing unit are aligned in a second direction

orthogonal to the first direction, wherein the first toner container and the second toner container are disposed closer to the first developing unit than to the second developing unit in the second direction, and wherein the pump unit is disposed so as to be overlapped with the first developing unit when viewed in a third direction perpendicular to both the first direction and the second direction.

14. The image forming apparatus according to claim 13, wherein the pump unit is disposed above the first developing unit.
